

100 PROBLEMATIC MCQS

O LEVEL PHYSICS (5054)

By Usman bin Aslam

- 1) A plastic cube, of density ρ , is compressed so that each of its sides becomes one-half of its original length.

What is the new density of the material of the cube?

- A 0.125ρ B 0.50ρ C 2.0ρ D 8.0ρ

- 2) The initial pressure of a volume V_1 of a fixed mass of gas is p_1 . The gas expands at a constant temperature until it has volume V_2 and pressure p_2 .

Which statement about the pressure and volume of the gas is **not** correct?

- A Pressure is measured in N/m^3 .
B Pressure multiplied by volume is a constant.
C Pressure is directly proportional to $\frac{1}{\text{volume}}$.
D The ratio of V_1 to V_2 is equal to the ratio of p_2 to p_1 .

- 3) A wave in the sea collides with a cliff.

A crest of the wave hits the cliff once every 6.0 s.

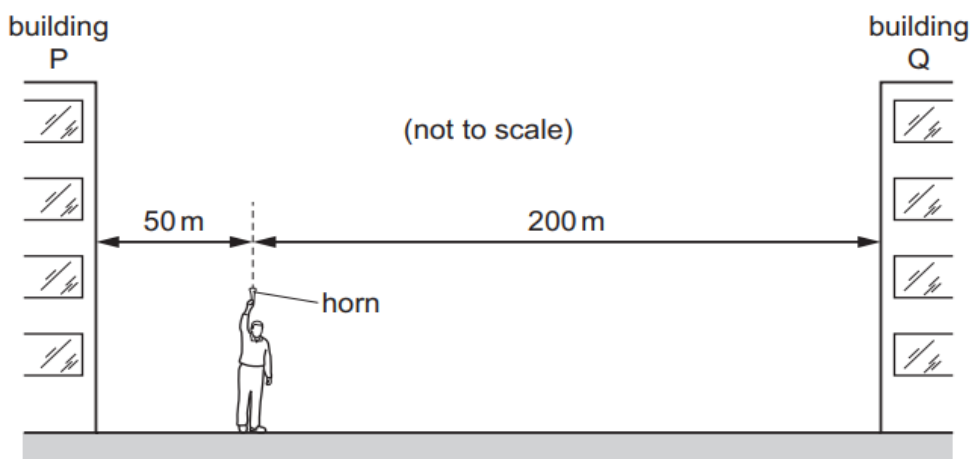
The horizontal distance between a crest and the adjacent trough of the wave is 4.5 m.

What is the speed of the wave?

- A 0.67 m/s B 0.75 m/s C 1.3 m/s D 1.5 m/s

- 4) A man stands between two tall buildings, P and Q.

He is 50 m from P and 200 m from Q.



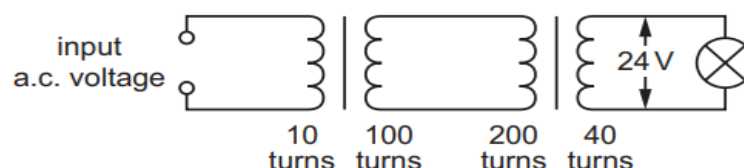
He sounds a horn.

He hears the first echo from building P and 1.0 s later he hears the first echo from building Q.

What is the speed of sound calculated using this information?

- A 250 m/s B 300 m/s C 330 m/s D 500 m/s

- 5) The diagram shows the circuit for a model power line with two transformers and a lamp at the output.



What is the input voltage?

- A** 0.12 V **B** 6.0 V **C** 12 V **D** 96 V

- 6) An elastic spring has an unstretched length of 30 cm.

A load of 6.0 N is hung from the spring and the length of the spring is now 66 cm.

The 6.0 N load is removed and the spring returns to its original length. A load of 2.0 N is now hung from the spring.

What is the new length of the spring?

- A** 22 cm **B** 40 cm **C** 42 cm **D** 52 cm

- 7) Charge of 0.40 C passes through a resistor in 1.0 s.

In 2.0 s, 20 J of thermal energy is produced in the resistor.

What is the potential difference across the resistor?

- A** 0.020 V **B** 0.040 V **C** 25 V **D** 50 V

- 8) Electrical power is transmitted by cables over long distances at very high voltages.

What are the effects of using a high voltage transmission system?

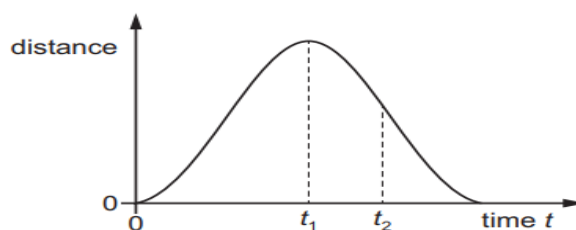
	power loss in the cables	current in the cables
A	high	high
B	high	low
C	low	high
D	low	low

- 9) At a depth d in sea-water, the total pressure experienced by a diver is $2P$, where P is atmospheric pressure.

At which depth is the pressure $4P$?

- A** $1.5d$ **B** $2d$ **C** $3d$ **D** $4d$

- 10) A train sets off from a station at time $t = 0$. The graph shows how the distance between the train and the station varies with time.



Which statement about the movement of the train between time t_1 and t_2 is correct?

- A Its speed is decreasing and it is moving away from the station.
 - B Its speed is decreasing and it is moving towards the station.
 - C Its speed is increasing and it is moving away from the station.
 - D Its speed is increasing and it is moving towards the station.
-

- 11) A 100 W lamp is switched on for five hours each day for three weeks.

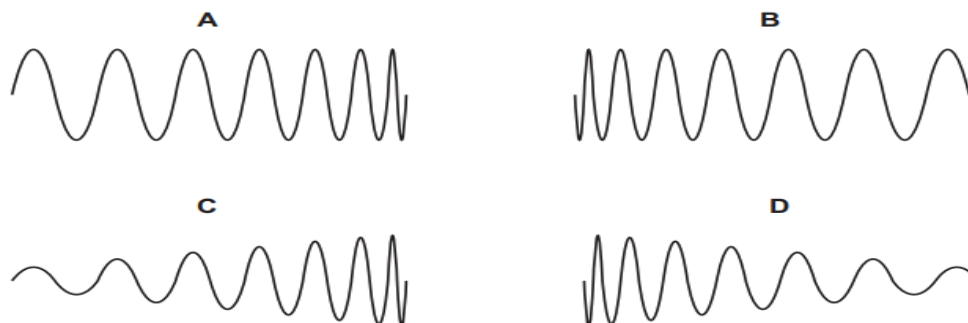
The cost of one unit of electricity is \$0.24.

How much does it cost to run the lamp for this time?

- A \$0.36 B \$0.84 C \$2.52 D \$25.20
-

- 12) In an alternating current (a.c.) generator, a magnet rotates near a coil of wire. The induced electromotive force (e.m.f.) in the coil is displayed on an oscilloscope screen.

Which trace is produced as the magnet slows down?

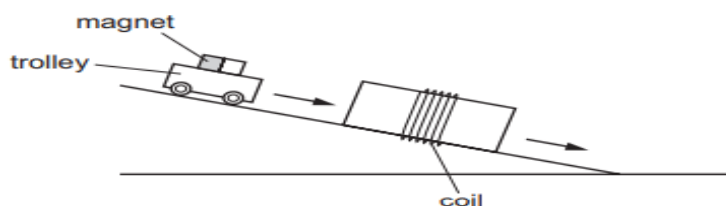


- 13) End X of a metal rod attracts the North pole of a compass needle.

Which statement about the rod is correct?

- A It is made of copper that is not initially magnetised.
- B It is made of copper with a South pole at X.
- C It is made of steel that is not initially magnetised.
- D It is made of steel with a North pole at X.

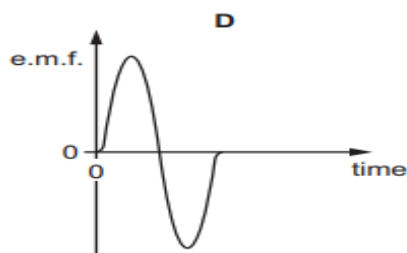
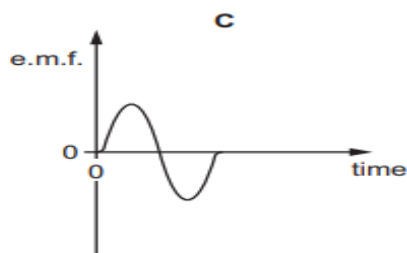
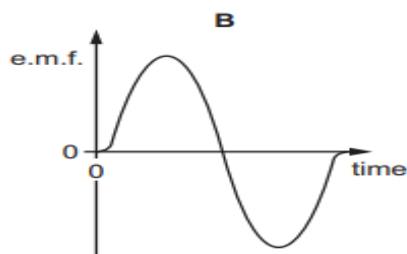
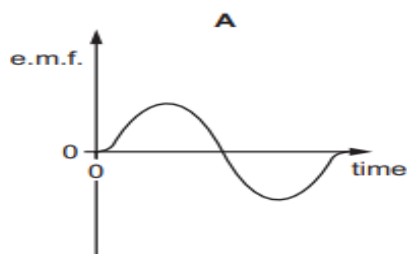
- 14) A trolley carrying a strong magnet rolls down a ramp at constant speed. It passes through a coil as shown.



An electromotive force (e.m.f.) is induced in the coil. A graph of the e.m.f. against time is plotted.

The experiment is repeated with different coils and with a steeper ramp. The trolley moves at a greater constant speed on the steeper ramp.

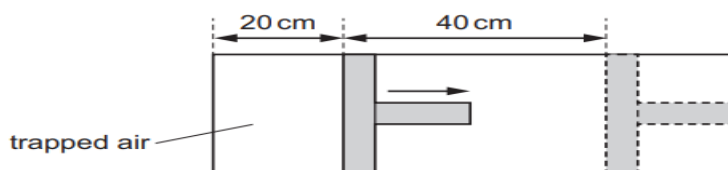
Which graph is produced using the coil with the least number of turns and the steepest ramp? All graphs are drawn to the same scale.



- 15) Air is trapped in a cylinder by a piston. The pressure of the air is p and the length of the air column is 20 cm.

The piston is moved outwards. The length of the air column increases by 40 cm.

The temperature of the air remains constant.



What is the new air pressure?

A $\frac{p}{2}$

B $\frac{p}{3}$

C $2p$

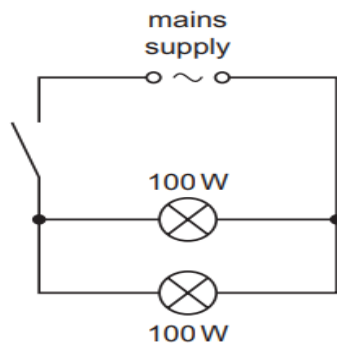
D $3p$

- 16) Technetium-99m is a radioactive isotope used in medical scanning. It is injected into the body and its emissions are detected outside the body.

Which characteristics of technetium-99m make it suitable for use in medical scanning?

- A** It has a long half-life and emits alpha radiation.
- B** It has a long half-life and emits gamma radiation.
- C** It has a short half-life and emits alpha radiation.
- D** It has a short half-life and emits gamma radiation.

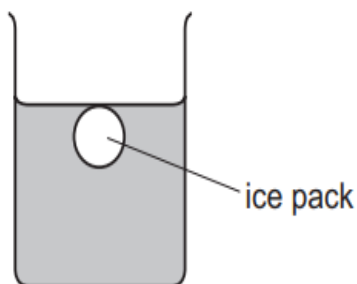
- 17) Two 100 W lamps are connected in parallel to the mains supply, as shown.



How much electrical energy is supplied by the mains when the switch is closed for 36 minutes?

- A** 0.030 kWh **B** 0.060 kWh **C** 0.12 kWh **D** 7.2 kWh

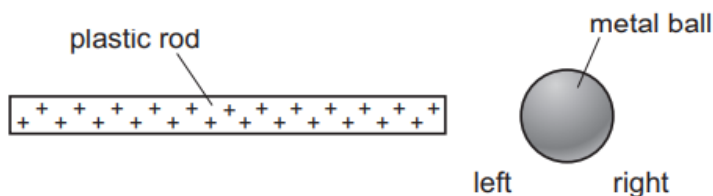
- 18) An ice pack is used to cool 0.25 kg of water. The specific heat capacity of water is $4.2 \text{ kJ}/(\text{kg } ^\circ\text{C})$.



How much thermal energy (heat) must the ice pack extract from the water to reduce the water temperature by 15°C ?

- A** 0.070 kJ **B** 1.1 kJ **C** 16 kJ **D** 250 kJ

- 19) A positively charged plastic rod is held near but not touching an insulated and previously uncharged metal ball. No charge leaves the rod.



What are possible charges on the left-hand side and right-hand side of the ball?

	charge on left-hand side of ball	charge on right-hand side of ball
A	$-2.0 \times 10^{-9} \text{ A}$	$-1.0 \times 10^{-9} \text{ A}$
B	$-2.0 \times 10^{-9} \text{ A}$	$+2.0 \times 10^{-9} \text{ A}$
C	$-2.0 \times 10^{-9} \text{ C}$	$+2.0 \times 10^{-9} \text{ C}$
D	$+2.0 \times 10^{-9} \text{ C}$	$-2.0 \times 10^{-9} \text{ C}$

- 20) Quantity X and quantity Y have the same unit Z.

X is the energy dissipated by a battery in driving unit charge round the complete circuit.

Y is the work done in a component in driving unit charge through the component.

What are X, Y and Z?

	X	Y	Z
A	electromotive force	potential difference	coulomb
B	electromotive force	potential difference	volt
C	potential difference	electromotive force	coulomb
D	potential difference	electromotive force	volt

- 21) A single metal bolt has a mass of 34 g.

Three of the bolts are immersed in a measuring cylinder that contains 30 cm^3 of water.

The reading on the measuring cylinder rises to 42 cm^3 .

What is the density of the metal?

- A** 0.81 g/cm^3 **B** 2.8 g/cm^3 **C** 7.5 g/cm^3 **D** 8.5 g/cm^3

- 22) A rocket of mass M when empty carries a mass M of fuel. The rocket and fuel travel at speed v . The engine of the rocket is fired and all of the fuel is expelled. The speed of the rocket increases to $2v$.

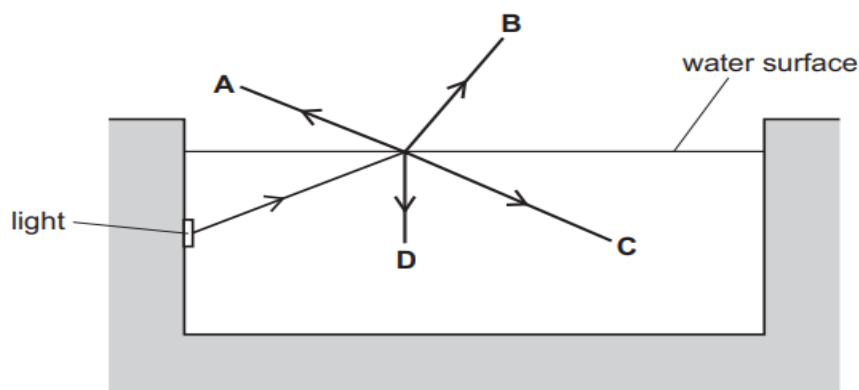
What happens to the kinetic energy of the rocket?

- A** It doubles.
B It halves.
C It increases by a factor of four.
D It stays the same.

- 23) A swimming pool is lit by an underwater light.

A ray of light is incident on the water surface.

What is the correct path for the ray of light?



- 24) $^{14}_6\text{C}$ is a natural radioactive isotope of carbon and is present in all living things. It has a half-life of 5700 years.

How old is a bone that is found to have only $\frac{1}{4}$ of the natural proportion of $^{14}_6\text{C}$?

- A 5700 years
- B 11 400 years
- C 17 100 years
- D 22 800 years

- 25) A kettle is connected to the 240 V mains supply using a plug containing a 13 A fuse.

The kettle contains water.

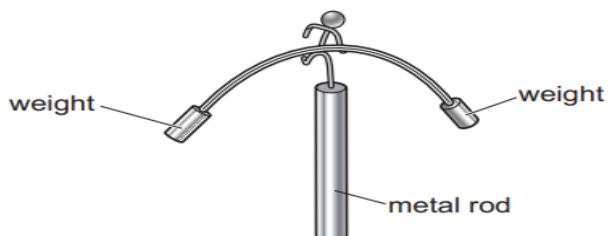
When it is switched on, the fuse blows. This happens again after a new fuse is fitted.

Someone replaces the fuse with a nail, and the kettle works.

What else might happen as a result of replacing the fuse with a nail?

- A A very large current overloads the wiring, causing a fire.
- B The kettle boils the water less quickly.
- C The kettle uses more energy to boil the water.
- D The water boils at a higher temperature.

- 26) A student makes a model toy which balances on the end of a metal rod.

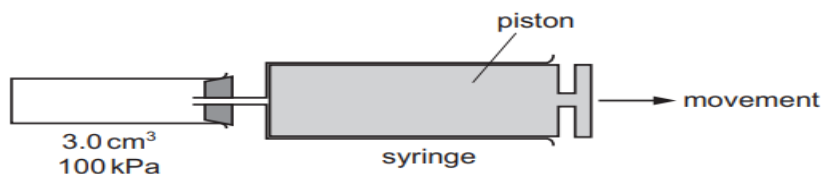


Why are the weights needed?

- A They increase the pressure on the rod.
- B They increase the weight of the model.
- C They lower the centre of mass below the top of the rod.
- D They raise the centre of mass above the top of the rod.

- 27) A small vessel of volume 3.0 cm^3 contains air at a pressure of 100 kPa.

The small vessel is connected to a syringe. The piston is fully inserted into the syringe.



The piston is moved slowly to the right so that the air in the syringe has a volume of 12 cm^3 . The temperature of the air does not change.

What is the pressure of the air in the syringe?

- A 20 kPa
- B 25 kPa
- C 80 kPa
- D 100 kPa

- 28) The 200 V battery of an electric car is charged with a current of 33 A for 4.0 hours. The efficiency of the charging process is 90%.

How much useable energy is transferred to the battery?

- A 21 MJ B 86 MJ C 95 MJ D 106 MJ

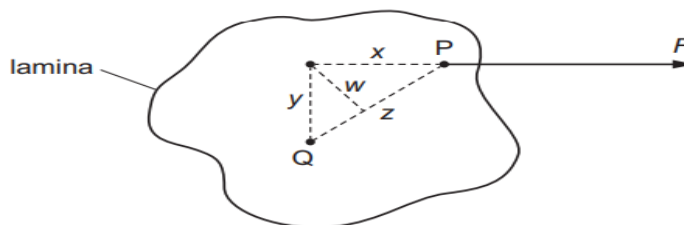
- 29) The thermal energy produced by an electric heater in three minutes is used to melt wax. The solid wax is initially at its melting point of 60°C .

The specific latent heat of the wax is 220 J/g . The heater supplies 7700 J of thermal energy to the wax. Some of the wax melts.

How much wax melts?

- A 0.58 g B 1.7 g C 12 g D 35 g

- 30) A length of thread is attached to a lamina at point P, as shown in the diagram.



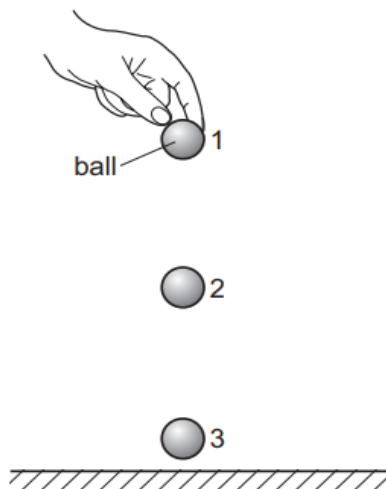
The lamina is free to rotate about point Q.

The tension in the thread is F .

What is the moment of F about Q?

- A Fw B Fx C Fy D Fz

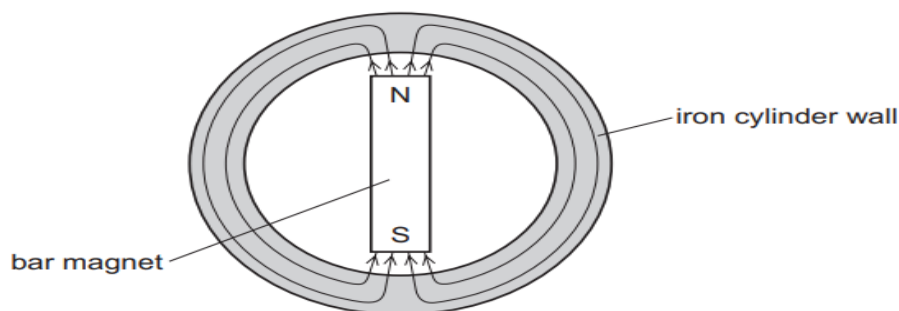
- 31) A ball is dropped from a height as shown.



Ignoring the effects of air resistance, which statement about the total energy of the ball is correct?

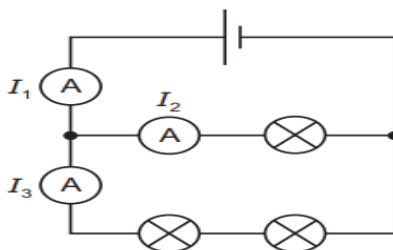
- A It is the same at all points. C It is greatest at point 2.
B It is greatest at point 1. D It is greatest at point 3.

- 32) A bar magnet is placed in a hollow iron cylinder. The diagram shows the magnetic field pattern produced.



What is the cause of the field inside the cylinder wall?

- A** electromagnetic generation **C** electrostatic induction
B electromagnetic induction **D** magnetic induction
-
- 33) Three identical lamps and three ammeters are connected as shown.



The readings on the ammeters are I_1 , I_2 and I_3 .

How are the readings related?

- A** $I_1 = I_2 = I_3$ **C** $I_1 > I_3 > I_2$
B $I_1 > I_2$ and $I_2 = I_3$ **D** $I_1 > I_2 > I_3$
-
- 34) A car of mass 800 kg has a forward acceleration of 2.5 m/s^2 . A frictional force of 1200 N opposes the motion of the car.

What is the driving force due to the engine of the car?

- A** 800 N **B** 1200 N **C** 2000 N **D** 3200 N
-
- 35) There is an astronaut on a planet which has a gravitational field strength one fifth of the value on Earth.

He hangs samples of rock from a spring balance designed for use on Earth. Each sample has a mass of 0.1 kg.

The spring balance reads 1.0 kg.

How many samples of rock are there?

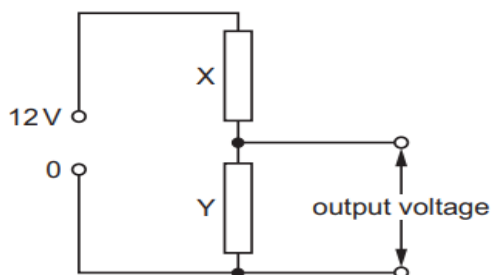
- A** 2 **B** 10 **C** 25 **D** 50

- 36)** A scientific research boat uses pulses of ultrasound to detect fish underneath it.
- A pulse of ultrasound is bounced from a group of fish 6000 m directly below the boat.
- The next pulse is sent out 2.0 s after the echo from the previous pulse has been received.
- The speed of ultrasound in water is 1500 m/s.
- What is the interval between the two pulses?
- A** 4.0 s **B** 6.0 s **C** 8.0 s **D** 10 s

- 37)** A 240 V mains circuit contains eight 60 W lamps in parallel. At the time when the lamps are switched on, the filaments are cold and the current is four times as large as the final steady current in the circuit.
- What is the initial current supplied by the mains?
- A** 0.25 A **B** 1.0 A **C** 2.0 A **D** 8.0 A

- 38)** A transformer has 4800 turns on its primary coil and 480 turns on its secondary coil.
- The primary coil is connected to a 240 V a.c. supply. The secondary coil is connected to a lamp.
- How does the output current in the lamp compare with the input current?
- A** higher frequency a.c.
B lower frequency a.c.
C current in one direction only
D the same frequency a.c.

- 39)** A potential divider uses a power supply of voltage 12 V. The resistors X and Y initially have equal resistances.

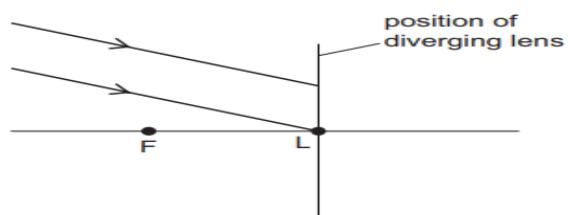


The resistance of X is halved.

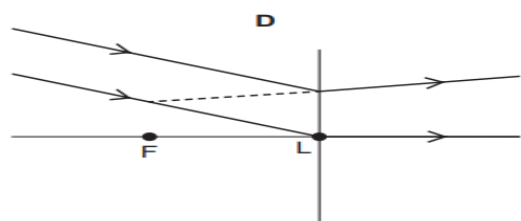
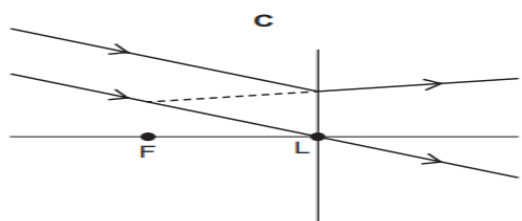
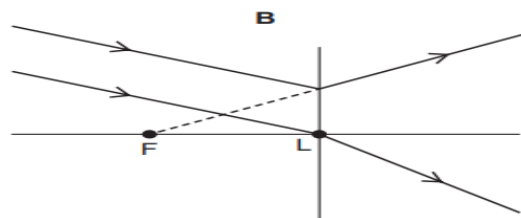
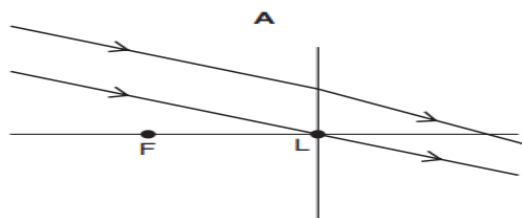
What is the change in the output voltage?

- A** -3.0 V **B** -2.0 V **C** +2.0 V **D** +3.0 V

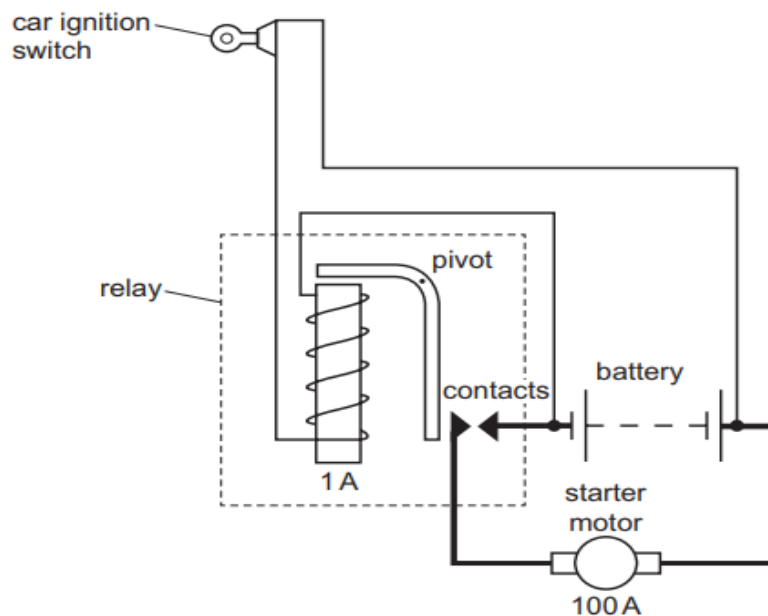
- 40) A parallel beam of light is incident on a thin diverging lens.
The focal length of the lens is FL, as shown in the diagram.



Which ray diagram shows the beam after it has passed through the lens?



- 41) The diagram shows a car ignition switch and starter motor.



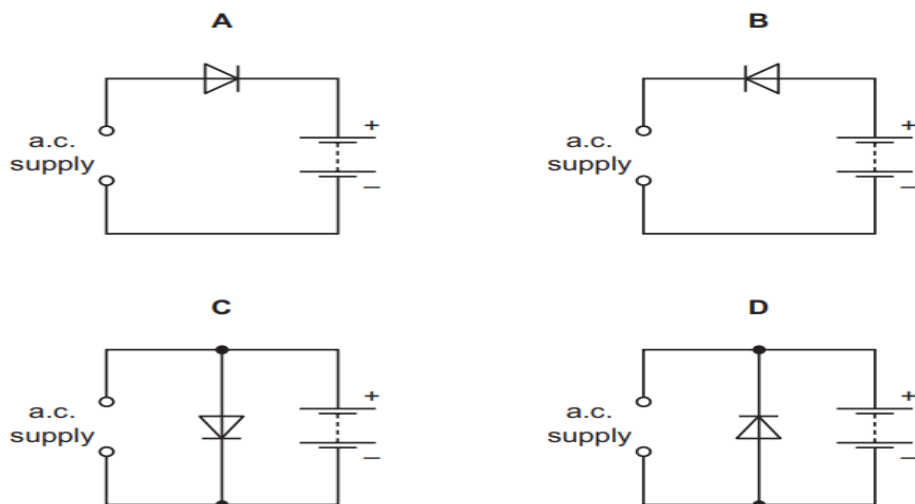
The ignition switch is in a circuit with long, thin wires. The starter motor is in a circuit with short, thick wires.

What is the explanation for the choice of wires?

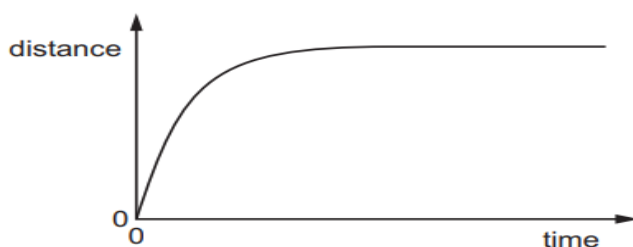
- A Each circuit needs to contain the same total mass of wire.
- B Thicker wires heat up more quickly when the relay is switched on.
- C Thin wires have lower resistances.
- D The ignition switch circuit carries a smaller current than the starter motor circuit.

- 42)** A car battery supplies a current in one direction. A current in the opposite direction recharges the battery.

Which circuit recharges the battery, using an alternating current (a.c.) supply and a diode?



- 43)** The diagram shows the distance–time graph for a moving object.



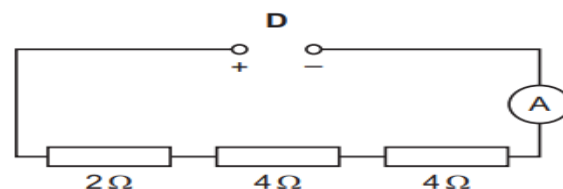
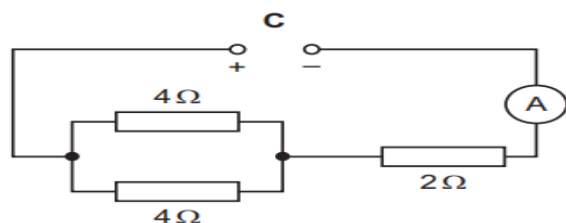
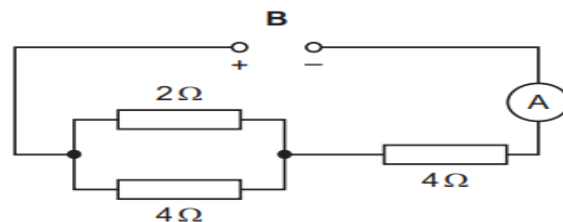
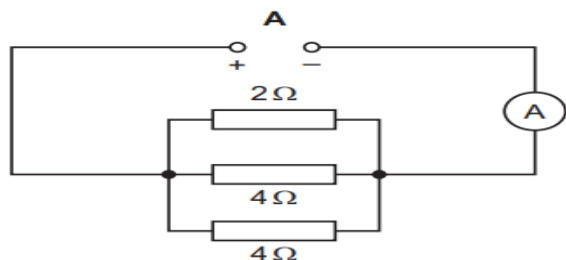
What is the moving object?

- A** a ball thrown vertically upwards and falling back to the thrower
 - B** a car, starting from rest, speeding up as it moves away from a traffic signal
 - C** a rock dropped from a high cliff, falling into the sea below
 - D** a train braking to a halt as it stops at a station
- 44)** In low-pressure hydrogen, there are no forces between the molecules except during collisions with each other.

Which statement describes the motion of a molecule between one collision and the next?

- A** It moves with changing speed and continuously changes direction.
- B** It moves with changing speed in a straight line.
- C** It moves with constant speed and continuously changes direction.
- D** It moves with constant speed in a straight line.

- 45) An ammeter is connected to three resistors and a power supply.
Which arrangement of resistors gives the greatest ammeter reading?



- 46) An isotope of lead, $^{212}_{82}\text{Pb}$, is radioactive. After three decays, two beta-particles and an alpha-particle have been emitted. Another radioactive isotope of lead is formed.

What is the mass number (nucleon number) of the new isotope of lead?

- A** 200 **B** 208 **C** 212 **D** 216

- 47) Each tyre of a car has an area of 100 cm^2 in contact with the ground.

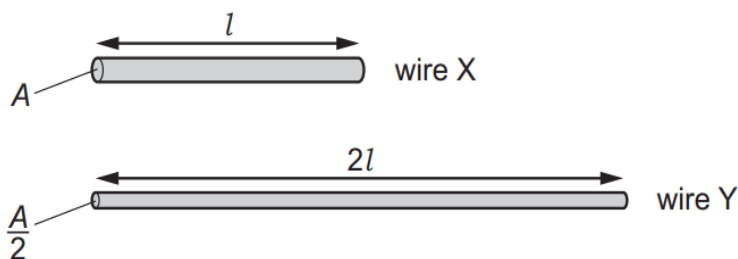
The car has a mass of 1600 kg. The weight of the car is equally distributed amongst the four tyres.

The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the ground?

- A** 4.0 N/cm^2 **B** 16 N/cm^2 **C** 40 N/cm^2 **D** 160 N/cm^2

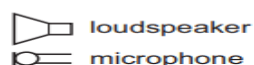
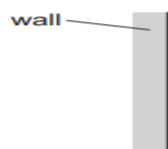
- 48) A copper wire X has resistance R . Another copper wire Y has twice the length and half the cross-sectional area of X.



What is the resistance of Y?

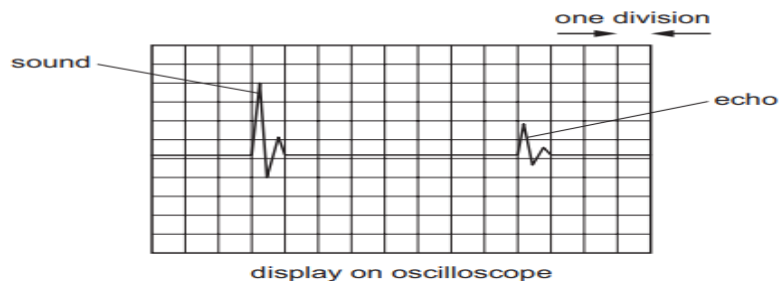
- A** $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$

- 49) A loudspeaker and a microphone are placed in front of a wall.



The loudspeaker makes a sound which is detected by the microphone.

The microphone is connected to an oscilloscope which is set so that each division on the screen represents 0.01 s. The microphone detects the original sound and the echo.



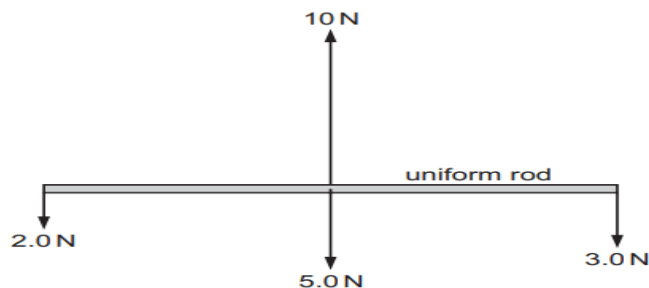
The speed of sound in air is 300 m/s.

What is the distance between the loudspeaker and the wall?

- A** 6.0 m **B** 12 m **C** 24 m **D** 48 m

- 50) A uniform rod of weight 5.0 N is held initially at rest.

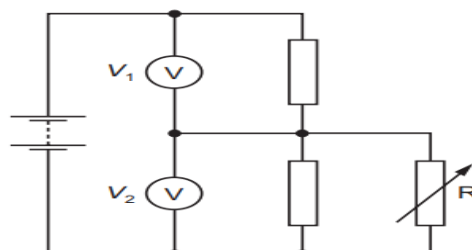
The diagram shows the forces acting on the rod when it is released.



What happens to the rod when it is released?

- A** It does not move. **C** It moves upwards.
B It moves to the right. **D** It starts to rotate.

- 51) The circuit diagram shows a variable resistor R connected in parallel to the lower half of a potential divider.

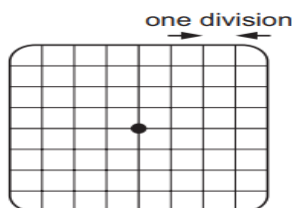


The resistance of R increases.

What happens to the two voltmeter readings?

	V_1	V_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

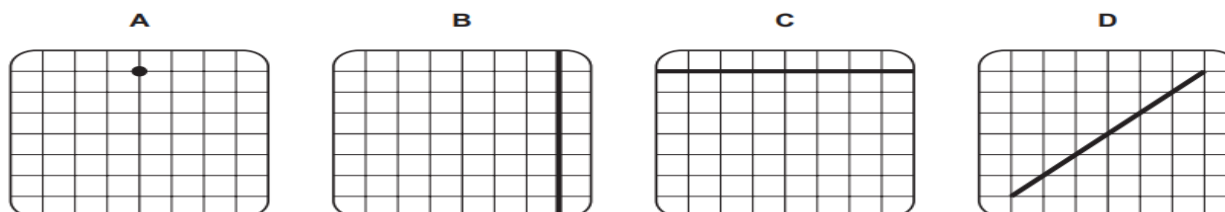
- 52) An oscilloscope is used to measure potential difference (p.d.). The trace with no input connected is shown.



A 1.5 V d.c. supply is connected to the oscilloscope.

The Y-gain is set at 0.5 V/div. The time-base is set at 0.5 ms/div.

Which trace shows a supply of 1.5 V d.c.?



- 53) A wire of length 0.50 m and cross-sectional area $1.0 \times 10^{-6} \text{ m}^2$ has a resistance of 0.75Ω .

Another wire of the same material has a length of 2.0 m and a cross-sectional area of $0.50 \times 10^{-6} \text{ m}^2$.

What is the resistance of the longer wire?

- A 0.094Ω B 0.38Ω C 1.5Ω D 6.0Ω

- 54) A rectangular block of metal has weight 6.0 N and measures $3.0 \text{ cm} \times 4.0 \text{ cm} \times 5.0 \text{ cm}$.

What is the smallest pressure that it can exert when resting on a horizontal surface?

- A 0.10 N/cm^2 B 0.30 N/cm^2 C 0.40 N/cm^2 D 0.50 N/cm^2

- 55) Which statement about a water wave is correct?

- A The amplitude is the vertical distance between a trough and a peak.
 B The frequency is the number of troughs passing a point in one second added to the number of peaks passing a point in one second.
 C The speed is the horizontal distance travelled per second by a peak.
 D The wavelength is the horizontal distance between a trough and a peak.

- 56) Ultrasound is used to map the ocean floor.

During one survey, the depth of water is 1200 m. An ultrasound pulse is sent from the surface and when it returns to the ship, another pulse is sent immediately. In any period of 8.0 s, five pulses are sent down from the surface.

What is the speed of the ultrasound in water?

- A 150 m/s B 300 m/s C 750 m/s D 1500 m/s

57) In a lighting circuit, the switch is placed in the live wire.

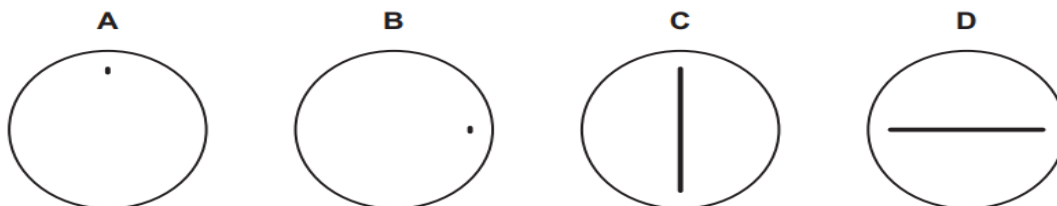
Why is this?

- A A lamp in the circuit can be isolated from the supply.
- B The fuse is in the neutral wire.
- C The switch does not work in the neutral wire.
- D Too much current flows in the earth wire.

58) An oscilloscope is connected to a d.c. power supply.

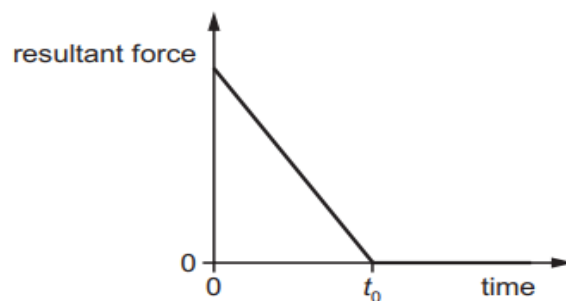
The time-base on the oscilloscope is switched off.

What is seen on the screen of the oscilloscope?

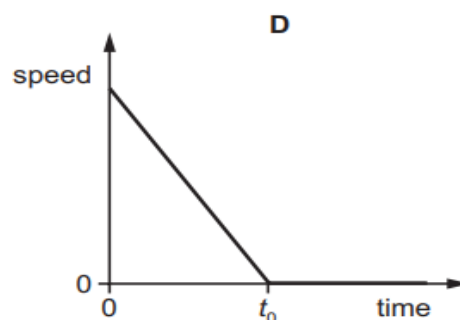
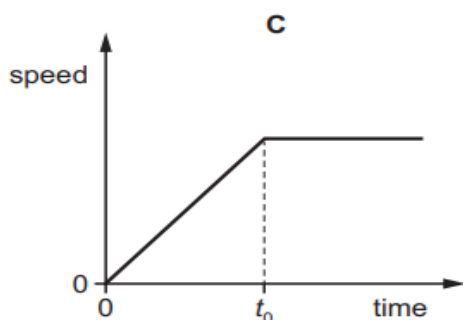
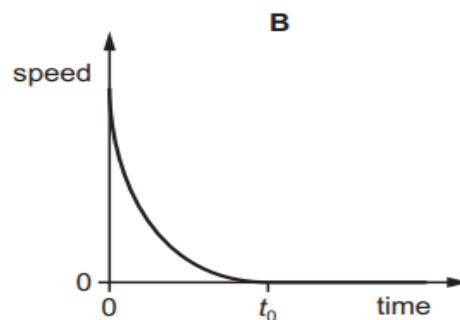
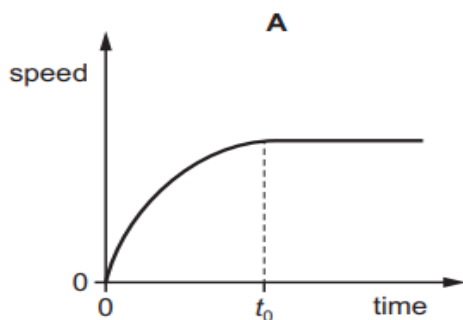


59) A resultant force acts on an object and causes it to move in a straight line.

The graph shows how the resultant force varies with time.



Which graph is the speed-time graph for the object?



- 60) During a thunderstorm, a lightning flash occurs when charge flows between a cloud and the Earth. On average, 20 C of charge flows during each flash and there are three flashes every minute.

What is the average current between the cloud and the Earth during the thunderstorm?

- A** 1.0 A **B** 36 A **C** 60 A **D** 1800 A
-

- 61) Why are batteries sometimes made from several identical cells arranged in parallel?

- A** The battery has a greater e.m.f. than a single cell on its own.
B The battery has a lower e.m.f. than a single cell on its own.
C The battery lasts longer than a single cell on its own.
D The battery supplies a smaller current than a single cell on its own.
-

- 62) A cylinder of conducting material has resistance R .

A second cylinder of the same material is twice as long but has half the cross-sectional area.

What is the resistance of the second cylinder?

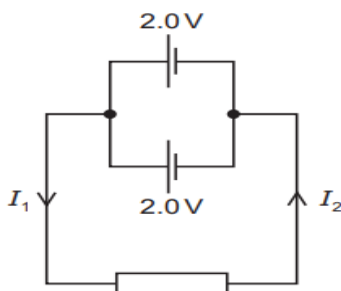
- A** $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$
-

- 63) During boiling and during melting, a substance either takes in energy or gives out energy.

Which row gives the energy transfer corresponding to the change of state?

	during boiling	during melting
A	gives out energy	gives out energy
B	gives out energy	takes in energy
C	takes in energy	gives out energy
D	takes in energy	takes in energy

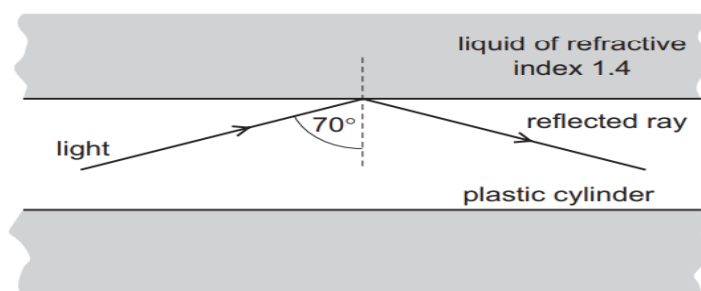
- 64) The battery in a circuit contains two identical cells connected in parallel.



Which row shows the relationship between currents I_1 and I_2 and gives the electromotive force (e.m.f.) of the battery?

	relationship	e.m.f. / V
A	$I_1 > I_2$	2.0
B	$I_1 > I_2$	4.0
C	$I_1 = I_2$	2.0
D	$I_1 = I_2$	4.0

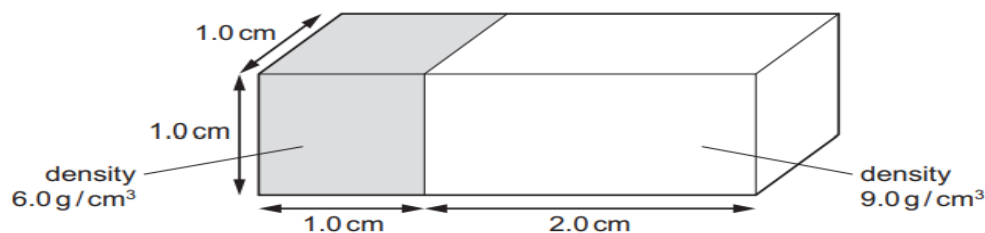
- 65) A solid plastic cylinder is immersed in a liquid of refractive index 1.4. Light travelling in the plastic cylinder strikes the inside surface at an angle of incidence of 70° . The light undergoes total internal reflection.



What are the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

- 66) Two blocks are joined together.

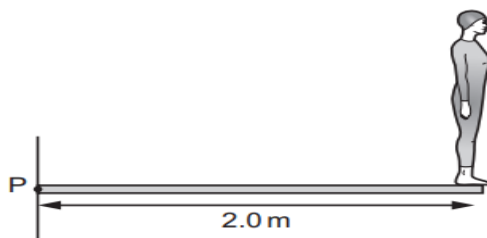


One block has a density of 6.0 g/cm^3 and the other has a density of 9.0 g/cm^3 .

What is the overall density of the two blocks joined together?

- A** 7.0 g/cm^3 **B** 7.5 g/cm^3 **C** 8.0 g/cm^3 **D** 15 g/cm^3

- 67) A diver of weight 500 N stands at the end of a springboard that is 2.0 m long and is fixed at point P.



The springboard has a weight of 500 N. The centre of mass of the springboard is in the centre of the board.

What is the total moment about point P of the diver and the board?

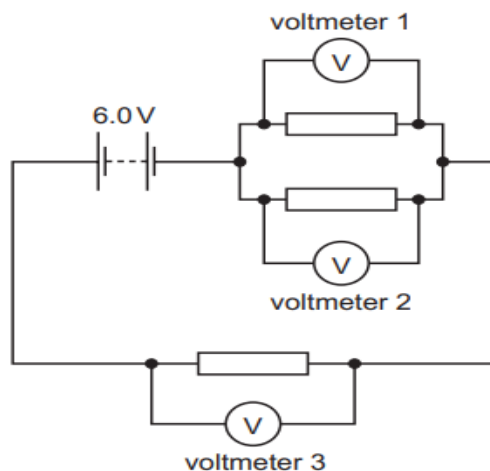
- A** 500 N m **B** 750 N m **C** 1000 N m **D** 1500 N m

- 68) An extremely large increase in pressure is needed to compress a liquid. A gas can be compressed by a much smaller increase in pressure.

Which statement explains this?

- A** Molecules repel each other very strongly when very close.
B The attractive forces between molecules are small at large distances.
C The molecules in the gas collide with the walls of their container and rebound.
D The molecules of a liquid are constantly moving at random.

- 69) A 6.0 V battery, three resistors and three voltmeters are connected in the circuit shown.



The reading on voltmeter 1 is 2.0 V.

What is the reading on voltmeter 2 and what is the reading on voltmeter 3?

	reading on voltmeter 2 /V	reading on voltmeter 3 /V
A	2.0	4.0
B	2.0	8.0
C	4.0	4.0
D	4.0	8.0

- 70) Water of depth 10m exerts a pressure equal to atmospheric pressure.

An air bubble rises to the surface of a lake which is 20m deep. When the bubble reaches the surface, its volume is 6.0 cm^3 .

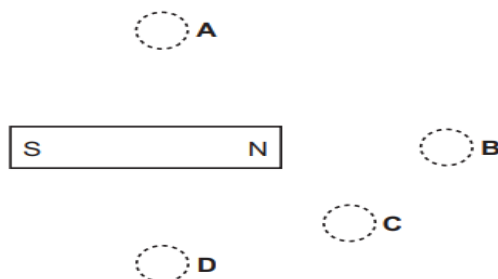
What is the volume of the air bubble at the bottom of the lake?

- A 2.0 cm^3 B 3.0 cm^3 C 12 cm^3 D 18 cm^3

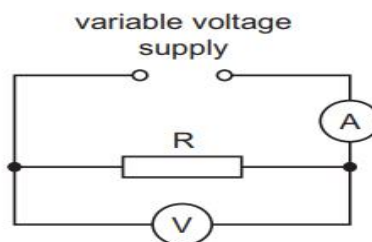
- 71) Four plotting compasses are placed near a bar magnet. You may ignore any effects of the Earth's magnetic field.

One compass appears like this  .

What is a possible position for this compass?



- 72) A student is investigating resistance using the circuit shown.



The resistance of R is approximately 5Ω .

What are the most suitable ranges for the voltmeter and for the ammeter?

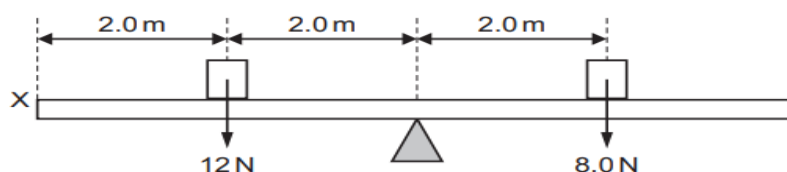
	voltmeter range	ammeter range
A	0–2 V	0–0.5 A
B	0–2 V	0–2 A
C	0–5 V	0–5 A
D	0–10 V	0–5 A

- 73) A sample of a radioactive isotope produces a count rate of 10 000 counts per minute. The half-life of the isotope is one day.

What was the count rate of the sample two days ago?

- A 2500 counts per minute
 B 5000 counts per minute
 C 20 000 counts per minute
 D 40 000 counts per minute

- 74) A uniform plank is pivoted at its mid-point. Two weights are added to the plank, one weight on each side of the pivot in the positions shown.

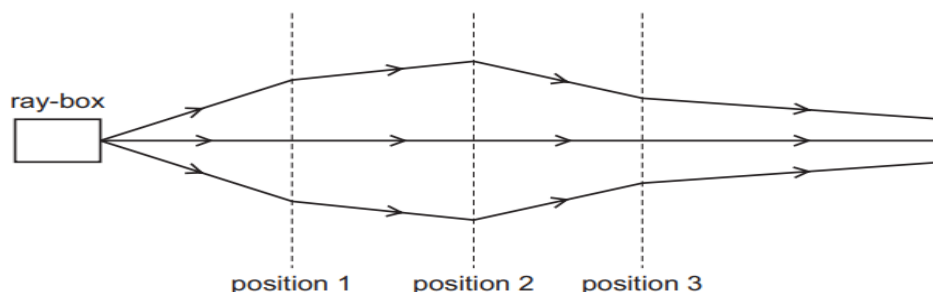


A vertical force is applied at point X to balance the plank.

What is the size and direction of this force?

	size / N	direction
A	2.0	downwards
B	2.0	upwards
C	4.0	downwards
D	4.0	upwards

- 75) The rays of light from a ray-box pass through three lenses placed at positions 1, 2 and 3.

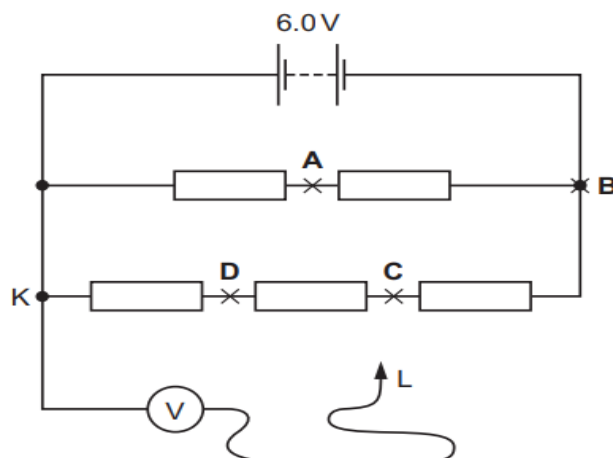


What type of lens is used at each position?

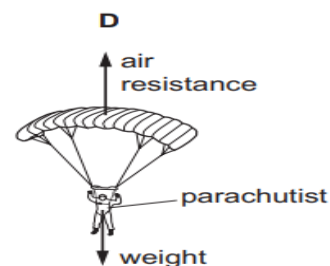
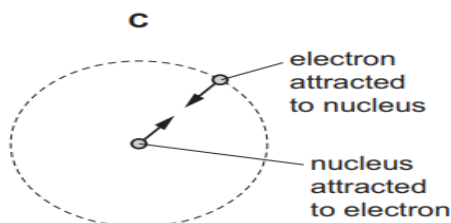
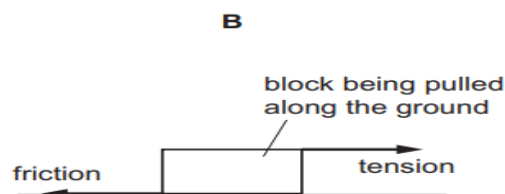
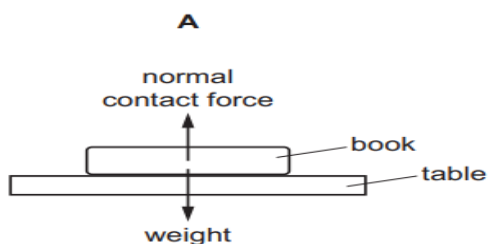
	position 1	position 2	position 3
A	converging	converging	converging
B	converging	converging	diverging
C	diverging	converging	diverging
D	diverging	diverging	converging

- 76) A 6.0 V battery is connected to a network containing five identical resistors. A voltmeter has one lead connected to point K as shown.

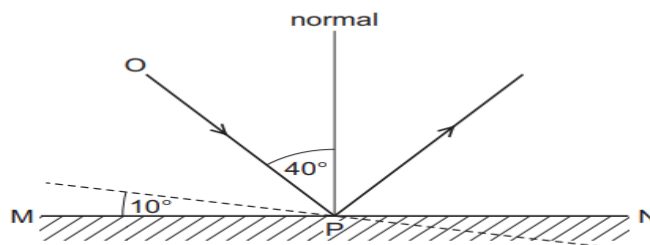
At which point should lead L be connected so that the voltmeter reads 3.0 V?



- 77) Each diagram shows two forces that are equal in magnitude.
Which diagram shows two forces that are a Newton's third law pair of forces?



- 78) The angle of incidence of ray OP on the plane mirror MN is 40° .

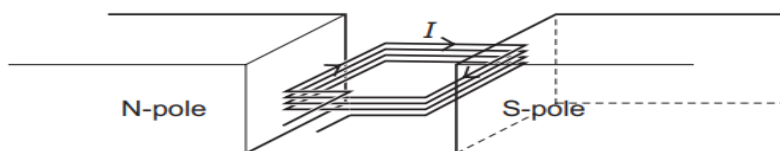


The mirror is rotated through 10° , as shown by the dashed line. The direction of the incident ray OP does not change.

What is the new angle of incidence?

- A** 30° **B** 40° **C** 50° **D** 60°

- 79) A coil P of N turns is made from a length L of wire. The coil carries a current I when between two magnetic poles.

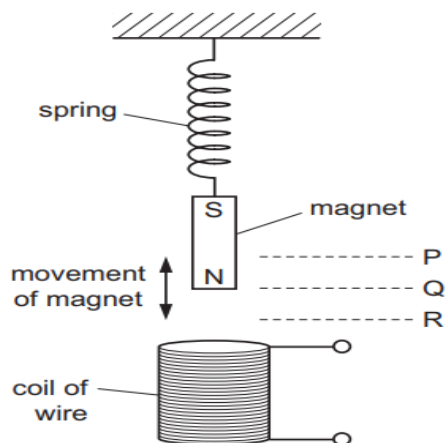


A similar coil Q of $2N$ turns is made from a length $2L$ of identical wire. It also carries a current I when between the two magnetic poles.

Which coil has the greater resistance and which coil experiences the greater turning effect?

	greater resistance	greater turning effect
A	P	P
B	P	Q
C	Q	P
D	Q	Q

- 80) A magnet moves up and down above a coil of wire.



The bottom of the magnet moves up and down between P and R.

Where is the bottom of the magnet when there is no induced electromotive force (e.m.f.) in the coil?

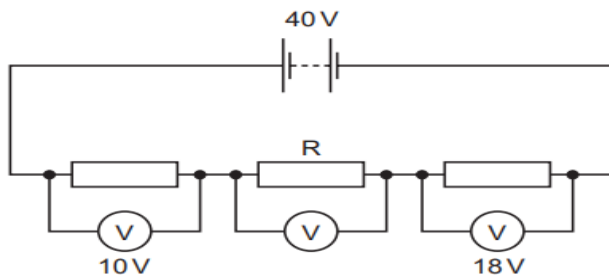
- A at P and at Q
- B at P and at R
- C at Q only
- D at R only

- 81) Two aluminium spheres P and Q, of masses 0.50 kg and 1.0 kg respectively, are released at the same time from the same height.

Why do they have the same acceleration?

- A The density of P is equal to the density of Q.
- B The force on P is equal to the force on Q.
- C The force on Q is double the force on P.
- D The gravitational field strength for Q is double that for P.

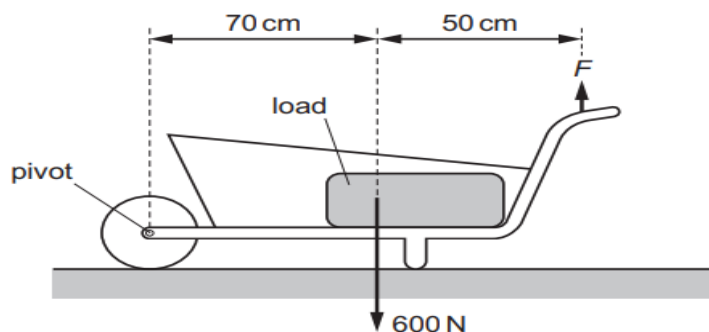
- 82) The circuit shows three resistors in series connected to a battery. Each resistor has a voltmeter across it and two of the voltages are shown.



What is the potential difference (p.d.) across the resistor R?

- A 12 V
- B 22 V
- C 30 V
- D 68 V

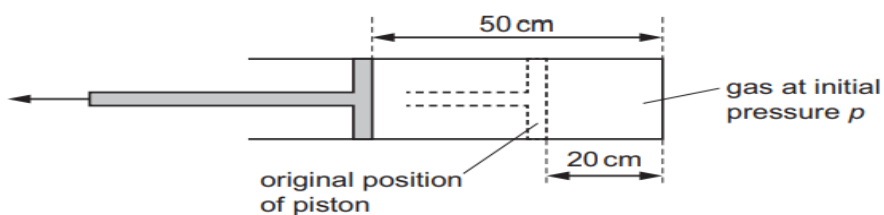
- 83) The total weight of the load and the wheelbarrow shown is 600 N.



What is the size of force F needed just to lift the loaded wheelbarrow?

- A 350 N B 430 N C 600 N D 840 N

- 84) A gas is trapped inside a cylinder by a movable piston. The length of the gas column is 20 cm and the pressure inside the cylinder is p .

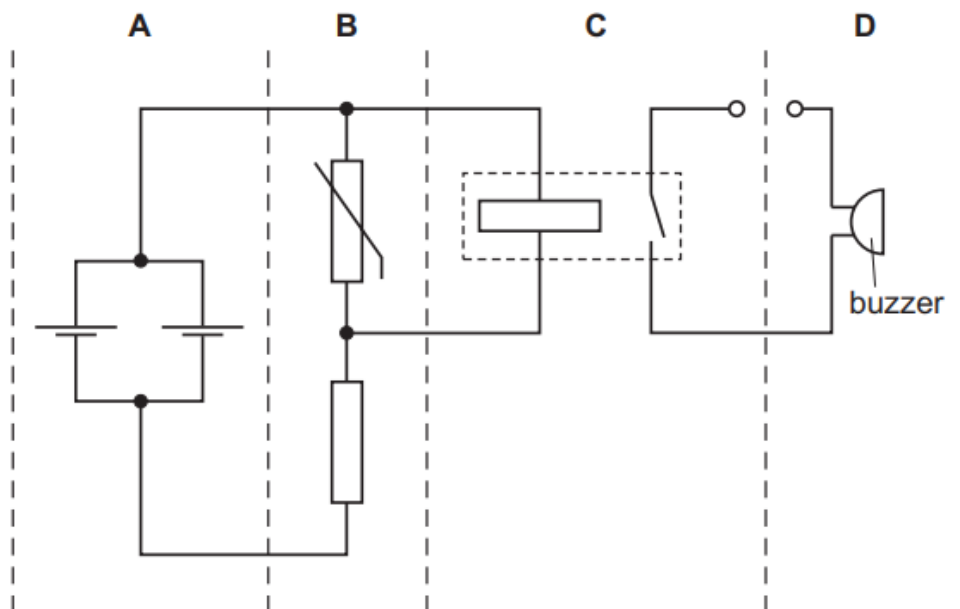


The piston is pulled out a distance of 30 cm, so that the length of the gas column is now 50 cm. The temperature of the gas does not change.

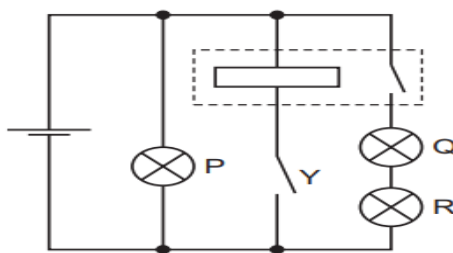
What is the new pressure of the gas?

- A $0.40p$ B $0.60p$ C $1.5p$ D $2.5p$

- 85) Which section of the circuit contains a potential divider?



- 86) In the circuit shown, all lamps are identical. Lamp P lights with normal brightness.

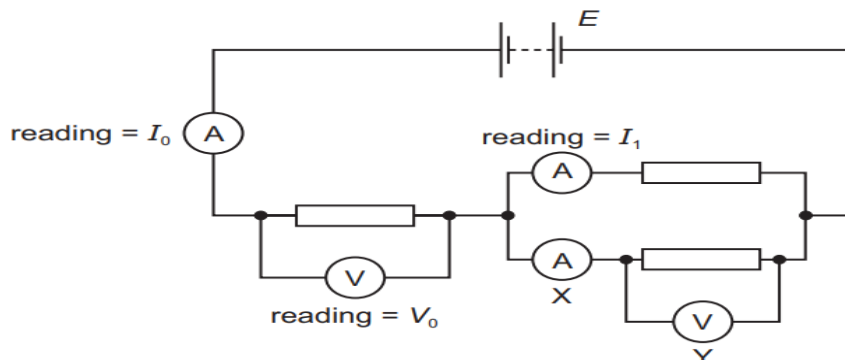


Switch Y is closed and lamps come on.

Which row indicates the brightness of the lamps?

	P	Q	R
A	dim	dim	dim
B	normal	dim	dim
C	normal	off	off
D	off	normal	normal

- 87) A battery of e.m.f. E is connected in a circuit containing three resistors.

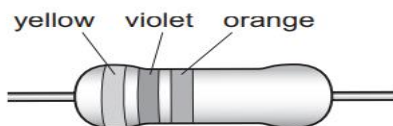


The readings of two ammeters and a voltmeter are shown.

Which readings are shown on ammeter X and on voltmeter Y?

	X	Y
A	$I_0 - I_1$	$E - V_0$
B	$I_0 - I_1$	$E + V_0$
C	$I_0 + I_1$	$E + V_0$
D	$I_0 + I_1$	$E - V_0$

- 88) A resistor is marked with three coloured bands.



black	brown	red	orange	yellow	green	blue	violet
0	1	2	3	4	5	6	7

Using the key provided, what is the resistance of the resistor?

- A** $374\ \Omega$ **B** $473\ \Omega$ **C** $47 \times 10^3\ \Omega$ **D** $37 \times 10^4\ \Omega$

- 89) A piece of wire has a resistance of $16\ \Omega$.

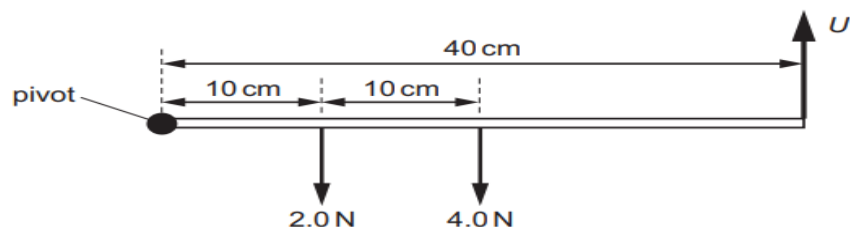
Another wire made from the same metal has four times the length and twice the cross-sectional area.

What is the resistance of the wire?

- A $8\ \Omega$ B $32\ \Omega$ C $96\ \Omega$ D $128\ \Omega$

- 90) A beam of length 40 cm is pivoted at one end.

The weight of the beam is $4.0\ \text{N}$ and acts at a point 20 cm from the pivot. A $2.0\ \text{N}$ weight hangs 10 cm from the pivot.

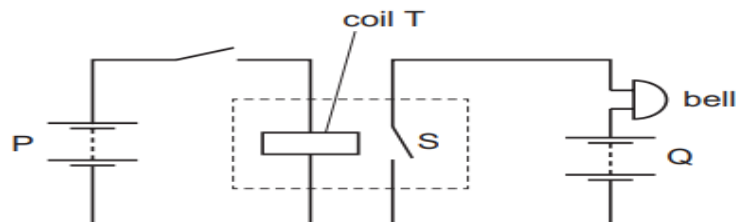


An upward force U is needed to keep the beam horizontal.

What is the size of U ?

- A $0.5\ \text{N}$ B $1.5\ \text{N}$ C $2.5\ \text{N}$ D $6.0\ \text{N}$

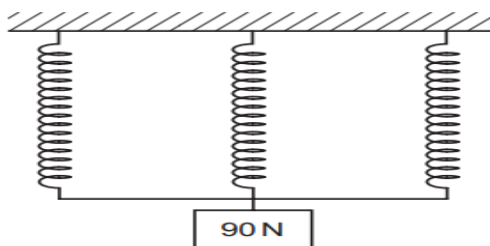
- 91) A relay is used in a circuit containing a bell.



How can the apparatus be altered to make the sound of the bell louder?

- A increase the number of turns on coil T
 B increase the voltage of battery P
 C increase the voltage of battery Q
 D move the coil closer to switch S

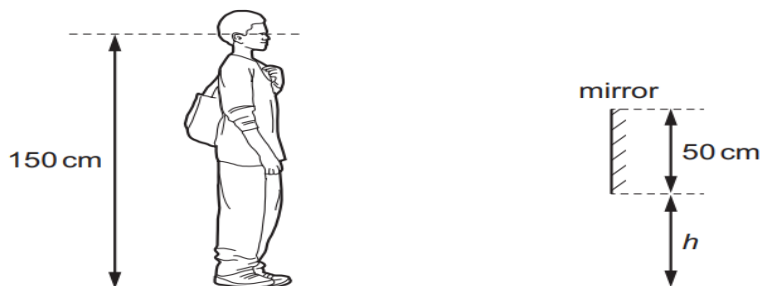
- 92) A spring extends by 6.0 cm when a $15\ \text{N}$ weight is suspended from its lower end.
 Three of these springs are used as shown to support a $90\ \text{N}$ weight.



What is the extension of each spring?

- A 2.0 cm B 12 cm C 36 cm D 110 cm

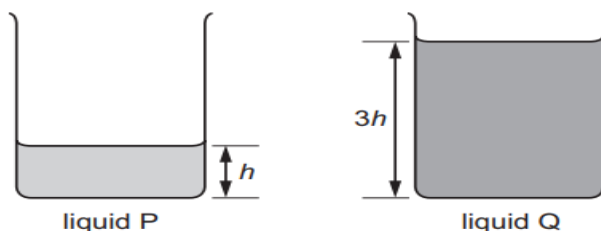
- 93) A shoe shop puts a mirror on the wall so that customers can look at their shoes. The length of the mirror is 50 cm. A customer has eyes 150 cm above ground level.



The bottom of the mirror is at height h above the ground.

What is the smallest value of h that allows the customer to see an image of his shoes in the mirror?

- 94) Two identical beakers contain the same mass of liquid. There is a different liquid in each beaker.



Liquid Q has a density ρ .

What is the density of liquid P?

- 95) Two leads emerging from a box are connected to a sensitive ammeter.

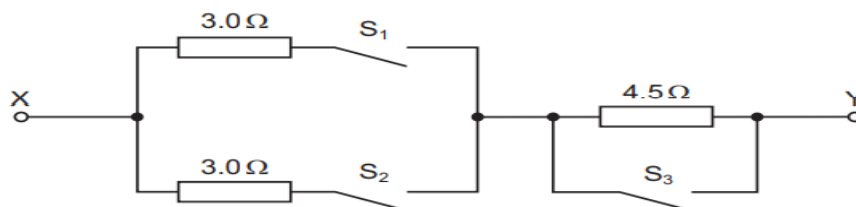


When a bar magnet moves towards the open end of the box, the needle of the ammeter deflects to the right. When the bar magnet stops, the needle returns to zero.

What is inside the box?

- A a coil alone
- B a coil connected in series with a cell
- C a light-dependent resistor (LDR) alone
- D an LDR in series with a cell

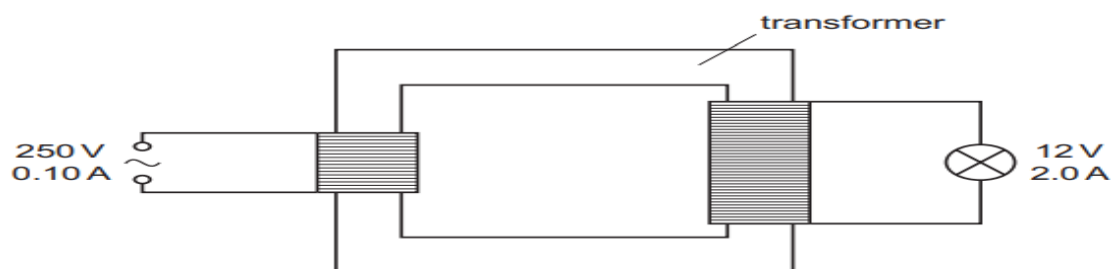
- 96) The diagram shows a circuit in which all the switches are open.



Which switch positions give a resistance of $6.0\ \Omega$ between X and Y?

	S_1	S_2	S_3
A	closed	closed	closed
B	closed	closed	open
C	closed	open	closed
D	closed	open	open

- 97) A transformer is used to operate a 12 V lamp from a 250 V mains supply.

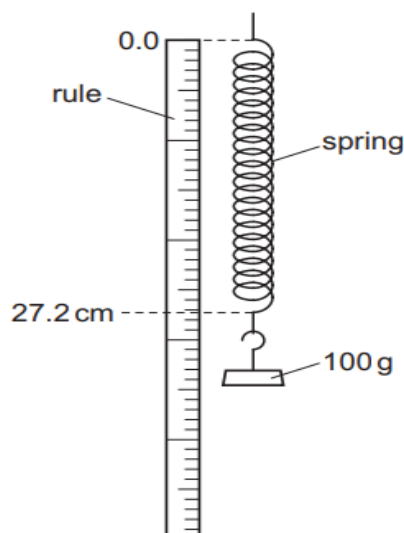


The mains current is 0.10 A . The current in the lamp is 2.0 A .

What is the efficiency of the transformer?

- A** 0.048 **B** 0.050 **C** 0.96 **D** 1.04

- 98) A 100 g mass is suspended from a spring next to a vertical metre rule. The top of the spring is level with the 0.0 cm mark. The bottom of the spring is level with the 27.2 cm mark.



The 100 g mass is replaced with a 600 g mass. The length of the spring is now 89.7 cm . The spring has not reached the limit of proportionality.

The 600 g mass is replaced with a 200 g mass. What is the length of the spring?

- A** 29.9 cm **B** 33.4 cm **C** 39.7 cm **D** 54.4 cm

- 99) A metal wire, of initial length 1000 mm, extends by 4 mm when a load of 2 N is added to it.

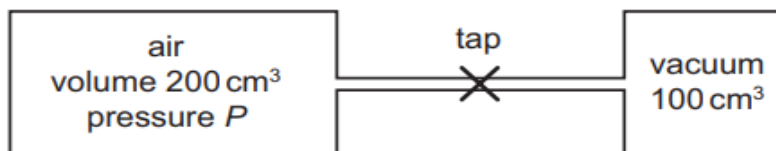
What is the length of the wire when a further 3 N is added, assuming that the wire does not extend beyond the limit of proportionality?

- A** 1006 mm **B** 1008 mm **C** 1010 mm **D** 1012 mm
-

- 100) Two cylinders are connected by a thin pipe.

One cylinder has a volume of 200 cm^3 and contains air at pressure P .

The other cylinder has a volume of 100 cm^3 and contains a vacuum (no air).



Initially, the tap is closed.

What is the final pressure of the air after the tap is opened?

- A** $\frac{P}{2}$ **B** $\frac{2P}{3}$ **C** $\frac{3P}{2}$ **D** $2P$